iReside

COMPREHENSIVE PROPERTY OPERATIONS & TECHNICAL ADMINISTRATION SUITE

COMPREHENSIVE SYSTEM OPERATIONS & TECHNICAL ADMINISTRATION MANUAL

Complete User Operations Guide, Financial Workflows & IT Technical Reference Handbook



DOCUMENT ID	DOC REF: IR-UNIFIED-2026-EN-V1.0
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Comprehensive Suite Architecture & Governance

DOCUMENT DIVISION	OPERATIONAL TARGET	PRIMARY DELIVERABLE
Part I: Operations	Landlords and Tenant Residents	Sequential workflows for daily property tasks.
Part II: IT Technical	DevOps and System Architects	Database schema, security invariants, and runbooks.
Pilot Implementation	Barangay Marulas, Valenzuela	Pilot compliance standards under RA 10173.

Prerequisites Before Getting Started

Ensure you have the following items ready before accessing iReside:

- **Smartphone or Computer:** Any modern Android phone, iPhone, or laptop.
- **Internet Connectivity:** Active Wi-Fi or mobile cellular data connection.
- **Valid Email Address:** Used for login verification and digital receipts.
- **Account Credentials:** Email and temporary password provided by management.

How to Sign In to Your Account

Follow these standard steps to authenticate into the system:

1. Open your browser and navigate to the official iReside portal.
2. Enter your registered email address in the **Email Address** field.
3. Enter your password in the **Password** field.
4. Click the **Sign In** button.
5. Retrieve the 6-digit verification code sent to your email inbox.
6. Type the code into **6-Digit Verification Code** and click **Verify & Sign In**.

Sign-In Credentials & Verification Matrix

CREDENTIAL FIELD	INPUT FORMAT	VERIFICATION RULE
Email Address	Standard valid email string	Must match active profile record.
Account Password	Minimum 8 alphanumeric characters	Case-sensitive credentials.
Verification Code	6-digit numeric OTP code	One-time code expiring in 10 minutes.

AUTHENTICATION STEP	USER ACTION	SYSTEM RESPONSE
1. Submit Details	Enter email and temporary password.	System validates registered account.
2. Receive Code	Retrieve 6-digit email passcode.	Code remains valid for ten minutes.
3. Authorize	Type code into verification prompt.	Redirects to role dashboard immediately.

> **Login Standard:** Two-factor verification protects accounts against unauthorized access.

> **Warning: Never share your password or verification code with anyone. Keep it confidential.**

How to Install the Mobile App

You can install iReside on your mobile phone for fast daily access:

ANDROID Installing on Android Phone

1. Open the left sidebar menu and click **Download App**.
2. Click **Download iReside App** to save the APK file.
3. Open your downloaded file and click **Install**.
4. When prompted by Android security, tap **Allow** to continue.

IPHONE Adding to iPhone Home Screen

1. Open the Safari browser and navigate to the iReside portal.
2. Tap the **Share** button at the bottom of the screen.
3. Scroll down and tap **Add to Home Screen**.
4. Tap **Add** in the top-right corner to complete setup.

Mobile Device Compatibility & Requirements

PLATFORM	MINIMUM SYSTEM	SUPPORTED FEATURES
Android Client	Android 8.0 (Oreo) or newer	Native APK wrapper, push alerts, camera uploads.
Apple iOS	iOS 14.0 or newer	Safari Home Screen PWA, camera uploads.
Desktop Browser	Chrome, Firefox, Safari, Edge	Full keyboard navigation, dual monitors, export tools.

The Emergency Spare Key (Account Recovery)

Just like a spare physical house key, this unlocks your account safely.

Emergency Spare Key Specifications

SECURITY PARAMETER	SPECIFICATION STANDARD	SAFETY DIRECTIVE
Code Structure	16 uppercase alphanumeric characters	Divided into 4 hyphenated blocks.
Storage Method	One-way bcrypt cryptographic hash	Plaintext key is never stored online.
Usage Restriction	Single-use emergency recovery token	Automatically rotated upon successful login.

> Store your Emergency Spare Key safely on a flash drive or notepad.

TASK T1 Sign Digital Lease Agreement (Sign Lease)

Before moving in, residents must digitally execute their lease contract:

1. Click **Leases** in the main navigation menu.
2. Click the **Review & sign** button at the top of the screen.
3. Carefully review monthly rent, security deposit terms, and building rules.
4. Click **Sign Document** to open the digital signature pad.
5. Draw your signature using your finger or mouse cursor.
6. Click **Finalize** and confirm to submit your signed lease.

TASK T2 Review Monthly Rent and Due Dates

Track your active rental dues, utility submeters, and payment history:

1. Click **Finance Hub** in the main navigation menu.
2. Review the top balance summary card labeled **Total Balance Due**.
3. Inspect current water and electric consumption calculated from submeters.
4. Review upcoming payment deadlines to avoid late payment notices.
5. Click **View Statement** to inspect an itemized billing breakdown.

Lease Terms & Invoice Status Matrix

TERM / STATUS	OPERATIONAL POLICY	TENANT DIRECTIVE
Monthly Rent	Payable on the first day of each month.	Submit payment receipts promptly via portal.
Security Deposit	Guarantees damages and unpaid bills.	Refunded within 15 days following move-out.
Unpaid / Overdue	Current or past due invoice balance.	Settle balance immediately to prevent penalties.
Paid / Pending	Approved or queued for verification.	Download and archive digital payment receipt.

Philippine Tenancy Compliance & Statutory Standards

REGULATORY STANDARD	STATUTORY REQUIREMENT	COMPLIANCE POLICY
Rent Control Act	Annual rent increase limits apply.	Automatic statutory rate capping.
Deposit Ceiling	Maximum two months advance rent.	Configurable deposit input constraints.
Grace Period	Standard 5-day payment leeway.	Penalties apply only after grace days.

TASK T3 Pay Rent via GCash or Maya

You can settle rental payments electronically without leaving home:

1. Open **Finance Hub** and click the **Upload Payment** button.
2. Scan the landlord GCash QR code or copy the recipient mobile number.
3. Complete payment inside GCash and capture a screenshot of your receipt.
4. Return to iReside and upload the payment proof screenshot image.
5. Type the exact **Reference Number** from your payment receipt.
6. Click **Submit Payment** to transmit verification to management.

SETTLEMENT STAGE	RESPONSIBLE ROLE	OPERATIONAL DELIVERABLE
1. Payment Submission	Tenant Resident	Uploads payment receipt screenshot with reference number.
2. Verification Queue	Accounting Ledger	Transfers invoice to pending review ledger queue.
3. Settlement Approval	Landlord Manager	Approves payment proof and issues digital receipt.

> **Payment Standard:** Double check the 13-digit GCash reference number before submitting verification.

TASK T4 Pay Cash Directly to Landlord

If you prefer paying rent with physical cash to management:

1. Visit the management office and submit the exact cash amount.
2. Wait for the landlord to record the transaction in iReside.
3. Open your Finance Hub screen and refresh the page.
4. Locate your new receipt row under **Payment History**.

Payment Method Comparison & Requirements

PAYMENT METHOD	VERIFICATION DATA	PROCESSING SLA
GCash / Maya Transfer	Screenshot and 13-digit reference number	Verified within 24 hours of upload.
Cash to Landlord	Handover to management office in person	Instant electronic ledger update on receipt.

TASK T5 Submit Repair Request (Report Repairs)

When experiencing damaged fixtures, plumbing issues, or electrical faults:

1. Click **Maintenance** in the main navigation menu.
2. Click the **+ Submit Request** button at the top right.
3. Select an issue category such as **Plumbing** or **Electrical**.
4. Type a clear, detailed description of the maintenance problem.
5. Click **Upload Photo** and choose clear picture evidence.
6. Click **Submit Ticket** to dispatch the repair work order.

REPAIR STAGE	RESPONSIBLE PARTY	OPERATIONAL STANDARD
1. Ticket Creation	Tenant Resident	Provides problem description and clear damage photos.
2. Triage & Schedule	Landlord Manager	Reviews urgency and assigns technician to work order.
3. Repair Completion	Assigned Technician	Fixes reported fault and verifies physical restoration.

> **Repair Standard:** Include well-lit close-up photos of damages to speed up repair scheduling.

TASK T6 Send Direct Message to Landlord

For inquiries, notices, or general questions with property management:

1. Click **Messages** in the left navigation sidebar.
2. Select the conversation thread with your property manager.
3. Type your question or inquiry in the message composer.
4. Click the **Send** icon to dispatch your message immediately.
5. You will receive a notification alert once management replies.

Maintenance Priority Matrix & SLA Times

FAULT CATEGORY	REPRESENTATIVE DAMAGE	TARGET RESOLUTION SLA
Plumbing	Water leaks, pipe burst, faucet failure	Resolved within 24 hours.
Electrical	Exposed wire, breaker trip, light failure	Resolved immediately or within 12 hours.
General Fixture	Window lock, door hinge, cabinet damage	Resolved within 48 to 72 hours.

TASK T7 Submit Move-Out Notice (Notice to Vacate)

When your lease concludes and you intend to vacate your unit:

1. Click **Leases** in the main navigation menu.
2. Review your active tenancy contract specifications.
3. Click the **Submit Move-Out Notice** button.
4. Select your intended move-out departure date on the calendar.
5. Provide your moving reason and click **Confirm Notice**.

TASK T8 Sign In Using Emergency Spare Key

If you lose your phone or cannot access your email inbox:

1. On the login page, click the **Use Emergency Key** link.
2. Enter your registered account email address.
3. Enter your unique 16-character Emergency Spare Key code.
4. Click the **Unlock Account** button to authenticate.
5. Set a new password and save your replacement emergency key.

Move-Out Inspection & Emergency Recovery Matrix

WORKFLOW AREA	REQUIRED ACTION	VERIFICATION STANDARD
Room Cleanliness	Trash removed and surfaces cleaned.	Cleaning fee charged if left dirty.
Key Handover	All gate and unit keys returned.	Missing keys deducted from deposit.
Final Submeters	Final water and electric readings taken.	Final usage deducted from deposit balance.
Spare Key Recovery	Input 16-character emergency code.	Resets password and rotates recovery key.

Security Deposit Settlement & Deduction Schedule

DEDUCTION ITEM	ASSESSMENT STANDARD	SETTLEMENT WINDOW
Unpaid Utilities	Final electric and water submeter readings.	Deducted directly from deposit balance.
Room Damage	Physical inspection and repair estimate.	Itemized repair receipts provided to tenant.
Remaining Balance	Deposit funds minus verified deductions.	Refunded within 15 days of departure.

> **Security Directive: Store your Emergency Spare Key offline. Do not share it over messages.**

TASK L1 Add New Rental Property and Units

Register a new rental building and its inventory in the portal:

1. Click **Properties** in the left navigation sidebar.
2. Click the **Add Property** button in the top action bar.
3. Enter your building title under **Property Name**.
4. Provide the complete address and select number of floors.
5. Specify room counts for each floor level.
6. Click **Create Property** to register the building in iReside.

ARCHITECTURAL PRESET	CORRIDOR CONFIGURATION	RECOMMENDED PROPERTY TYPE
Double Loaded	Central hallway with rooms on both sides.	Multi-story apartment and boarding buildings.
Single Loaded	Single corridor with open exterior railing.	Open-air rental complexes and row houses.
U-Shape Layout	Three-sided building enclosing central courtyard.	Multi-wing gated compound residential rental units.

> **Unit Map Standard:** Drag room cards along the interactive grid to match corridors.

TASK L2 Configure Interactive Unit Map

Create visual 2D floorplans for effortless hallway oversight:

1. Click **Unit Map** in the main navigation menu.
2. Choose an architectural preset like **Double Loaded** or **Single Loaded**.
3. Drag and drop room cards to match your corridor layout.
4. Click any unit card to edit numbers and monthly rates.
5. Click **Save Layout** to publish your floor plan.

Property Specifications & Boundary Rules

CONFIGURATION FIELD	INPUT CONSTRAINT	OPERATIONAL GUIDANCE
Property Title	Maximum 60 alphanumeric characters	Use distinct building name for recognition.
Complete Address	Barangay Marulas street location	Must be physical location within pilot area.
Floor Count	Number from 1 to 10 floors	Generates structural floor plan levels.

TASK L3 Configure Utility Rates & Payment Details

Establish utility billing tariffs before calculating water and power:

1. Click **Utility Billing** in the left navigation menu.
2. Open the **Electricity & Water Rates** configuration tab.
3. Set electricity rate per kWh and water rate per cubic meter.
4. Enter your official **GCash Mobile Number** and account name.
5. Click **Save Financial Configuration** to apply rates across units.

> Tip: Ensure your GCash account name matches official payment recipient details.

TASK L4 Add New Resident Tenant (Quick Add)

The fastest onboarding method when meeting residents in person:

1. Click **Tenants** in the sidebar and click **+ Add Tenant**.
2. Select the **Quick Add** onboarding tab.
3. Enter the resident Full Name, Email Address, and Phone Number.
4. Assign a vacant room, monthly rent, and lease start date.
5. Click **Create Tenant Account** to generate login credentials.

Utility Tariff Structure & Billing Specifications

UTILITY SERVICE	MEASUREMENT UNIT	BILLING FORMULA
Electricity	Kilowatt-hour (kWh)	Usage multiplied by configured rate per kWh.
Water	Cubic meter (m^3)	Usage multiplied by configured rate per cubic meter.
GCash Account	11-digit mobile number	Displayed directly on tenant invoices.

Tenant Onboarding Options & Account Provisioning

ONBOARDING OPTION	INTENDED APPLICANT	SYSTEM OUTPUT
Quick Add	Walk-in tenants meeting in person.	Instant account with temporary password.
Self-Onboarding	Remote applicants using mobile phones.	Secure application link sent via SMS or chat.
Digital Lease	Approved incoming residents.	Binding electronic contract ready to sign.

> **Privacy Standard:** Tenant phone numbers and contact details remain protected under Republic Act 10173.

TASK L5 Generate Self-Onboarding Invite Link

Allow prospective tenants to complete application forms on mobile:

1. Inside the Add Tenant dialog, click **Self-Onboarding Link**.
2. Select the target vacant unit and link expiration timeframe.
3. Click **Generate Onboarding Link** to create the registration URL.
4. Click **Copy Link** or show the QR code to the applicant.
5. You receive an instant alert when the tenant submits their application.

ONBOARDING CHANNEL	WORKFLOW METHOD	PRIMARY BENEFIT
Quick Add Form	Landlord enters resident details directly.	Instant credentials created for walk-in residents.
Self-Onboarding Link	Applicant completes mobile registration form.	Eliminates manual paperwork during remote leasing.
QR Code Poster	Prospective tenant scans physical sign.	Instant access during on-site property walkthroughs.

> **Onboarding Standard:** Verify government ID photos and mobile numbers before executing contracts.

TASK L6 Review and Process Walk-in Applications

Process rental applicants who visit your management office:

1. Open **Tenants** in the sidebar and choose the **Applications** tab.
2. Click the applicant name to inspect uploaded identity credentials.
3. If documentation is approved, click the **Approve Application** button.
4. Select the unit assignment and generate the digital lease agreement.
5. The platform automatically sends sign-in credentials to their verified email.

Applicant Verification & Document Review Criteria

VERIFICATION ITEM	REQUIRED EVIDENCE	APPROVAL DIRECTIVE
Government ID	UMID, Passport, or Driver License	Verify photo matches applicant name clearly.
Mobile Contact	11-digit Philippine mobile number	Perform test phone call or SMS inquiry.
Assigned Room	Vacant unit on interactive map	Confirm room status before lease generation.

TASK L7 Verify and Settle Tenant Payments

Confirm electronic GCash transfers or direct cash collections:

1. Click **Finance Hub** in the left navigation sidebar.
2. Under **Pending Verification**, click the submitted tenant payment row.
3. Review the uploaded receipt screenshot and match the reference number.
4. Click the **Confirm Payment** button to approve the transaction.
5. The tenant balance updates instantly and a receipt is issued.

TASK L8 Record Submeter Readings (Electric & Water)

Automate tenant utility billing calculations effortlessly:

1. Click **Utility Billing** and open the **Submeters** tab.
2. Select a room and enter current numbers into **Current Reading**.
3. The system computes exact consumption from previous recorded readings.
4. Total charges are calculated using your configured utility tariff rates.
5. Click **Save Reading & Generate Bill** to post charges.

Payment Settlement & Submeter Invoicing Matrix

WORKFLOW ACTION	VERIFICATION DATA	LEDGER RESULT
GCash / Maya	Match 13-digit reference number against statement.	Mark invoice paid and issue receipt.
Physical Cash	Confirm exact cash received in office.	Record cash transaction in ledger.
Partial Payment	Accepted payment is less than total due.	Deduct paid amount and retain remaining balance.
Submeter Billing	Net usage multiplied by tariff rate.	Consolidated invoice posted to tenant portal.

Utility Tariff Standards & Meter Variance Policies

UTILITY SERVICE	BILLING BASELINE	AUDIT VERIFICATION STANDARD
Electricity (kWh)	Submeter consumption multiplied by tariff rate.	Monthly photo documentation of analog dials.
Water (Cubic Meters)	Net volume usage times configured rate.	Checked against master water utility bill.

> **Audit Standard:** Take reference photos of analog meters during monthly inspections to verify records.

TASK L9 Manage and Resolve Maintenance Requests

Process repair work orders submitted by building residents:

1. Click **Maintenance** in the left navigation sidebar.
2. Click any ticket marked with a red tag or **Pending** status.
3. Review the problem description and photo evidence submitted by resident.
4. Assign the repair: **Landlord Repair** or **Third Party Repair**.
5. Click the **To Process** button to start scheduling.
6. Once work is finished, click the **Mark as Completed** button.

ADVISORY TYPE	BROADCAST SCOPE	PRIORITY INDICATOR
Utility Interruption	Scheduled water or electrical repair outages.	Urgent banner on tenant dashboard.
Payment Reminder	Upcoming monthly rent settlement deadlines.	Standard notification alert feed.
Building Advisory	Waste disposal schedules and quiet hours.	Community bulletin post on board.

> **Communication Standard:** Published announcements broadcast instantly across all active resident mobile devices.

TASK L10 Post Building Announcements

Broadcast essential building advisories (such as water maintenance):

1. Click **Community Hub** in the left navigation sidebar.
2. In the composer toolbar, select the **Announcement** tab.
3. Type the full notice details for your building residents.
4. Click **Photo** if you wish to attach an informative image.
5. Click the **Publish** button in the bottom right corner.

Maintenance Workflow & Resolution Protocol

WORKFLOW STAGE	RESPONSIBLE ACTOR	TARGET RESOLUTION STANDARD
Initial Review	Landlord Property Manager	Review damage photo within 12 hours.
Work Assignment	Internal Caretaker or Vendor	Schedule physical repair date with tenant.
Final Sign-Off	Landlord and Resident Tenant	Confirm repairs and close maintenance ticket.

TASK L11 Process Move-Out and Security Deposit

Execute structured move-out settlements when residents vacate:

1. Click **Leases** in the sidebar and open **Move-Out Requests**.
2. Complete the physical room inspection using the built-in checklist.
3. Enter itemized deductions for damages or outstanding utility balances.
4. The system calculates the net refundable **Security Deposit** amount.
5. Click **Finalize Move-Out** and disburse remaining funds to tenant.
6. The assigned room status returns to **Vacant** automatically.

> Tip: The system automatically deducts unpaid utility bills from deposits.

TASK L12 Export Financial Reports and Ledgers

Extract verified income and expense records for financial auditing:

1. Click **Finance Hub** in the menu and select **Reports**.
2. Select your reporting period by month, quarter, or annual range.
3. Inspect total collected rent, electricity fees, and water charges.
4. Click the **Export PDF** button to download a formatted statement.
5. Save or print the document for official property bookkeeping records.

Move-Out Settlement & Security Deposit Deductions

DEDUCTION ITEM	ASSESSMENT STANDARD	SETTLEMENT DIRECTIVE
Utility Arrears	Final submeter meter readings.	Deducted directly from deposit balance.
Room Damages	Damage exceeding normal wear and tear.	Itemized repair receipts provided to tenant.
Net Refund	Remaining security deposit balance.	Transferred via GCash or cash within 15 days.

Financial Reporting & Tax Audit Export Reference

REPORT FORMAT	COVERAGE PERIOD	KEY FINANCIAL METRICS
Monthly Rent Ledger	Current billing month	Total collected rent, unpaid arrears, utility fees.
Annual Revenue Sheet	12-month calendar year	Gross rental collections, deductions, net income.
Submeter Utility Audit	Monthly meter readings	Initial reading, final reading, total billable units.

> **Bookkeeping Directive:** Retain exported PDF financial reports for 5 years for tax compliance.

Troubleshooting Frequently Encountered Issues

1. Did not receive 2FA verification email?
Check your Spam or Junk folder. Wait 60 seconds before clicking Resend Code.

2. Forgotten password and unable to access email?
Use your Emergency Spare Key. Click Use Emergency Key on the login screen.

3. Weak internet connection while submitting payments?
Save your receipt screenshot. Upload proof once your internet connection is restored.

4. Submitted payment not yet deducted from balance?
Landlords must review and confirm receipts before balances update on screen.

5. How to upload clear repair request photos?
Turn on room lighting. Capture close-up photos of the specific damaged fixtures.

6. Who to contact during building emergencies?
Contact management emergency hotlines or Barangay Marulas Rescue Services immediately.

7. How to resolve camera permission errors when uploading receipts?
Open phone browser settings. Tap Site Settings, select Camera, and choose Allow.

Data Privacy & Legal Compliance Standards

REGULATORY MANDATE	PROTECTED DOMAIN	COMPLIANCE IMPLEMENTATION
RA 10173 (Data Privacy)	Tenant personal information	Encrypted storage of ID photos and phone numbers.
RA 8792 (E-Commerce)	Digital lease agreements	Legally binding electronic signatures on contracts.
Audit Trail Standard	Financial payment records	Immutable transaction timestamps on receipts.

Essential Security Guidelines

- **Keep Passwords Secret:** Never share passwords with anyone. Use mixed letters and numbers.
- **Store Emergency Keys Safely:** Save your spare key on a flash drive or notebook.
- **Beware of Phishing Messages:** Never log in through text message links. Use official URLs.
- **Archive Payment Receipts:** Save reference numbers and keep digital payment proof files.
- **Protect Personal Identity Data:** Government IDs remain encrypted and restricted to verified management.

Frequently Asked Questions (FAQs)

Q: How are monthly electricity and water bills computed?
System subtracts previous meter readings from current readings and multiplies tariffs.

Q: Are digital lease signatures legally binding in the Philippines?
Yes. Electronic signatures comply with Philippine Republic Act No. 8792.

Q: How is resident personal data protected inside iReside?
System complies with the Philippine Data Privacy Act of 2012 (RA 10173).

Q: What should I do if my GCash payment fails during transfer?
Wait 5 minutes, verify GCash wallet balance, and re-attempt payment transfer.

Official Helpline & Technical Support

For operational support, reach out to: **support@ireside.ph** | Tel: **(02) 8292-1000** (Monday through Saturday, 8:00 AM – 5:00 PM). Barangay Marulas Hall, MacArthur Highway, Valenzuela City.

Local Emergency & Assistance Hotlines

PUBLIC AGENCY	SERVICE SCOPE	EMERGENCY CONTACT
Barangay Hall	Barangay Marulas Community Desk	(02) 8292-1000
Fire Station	Valenzuela Fire Protection Service	160 / (02) 8292-2222
Police Substation	Marulas Police Community Precinct	166 / (02) 8292-1111

Operational Glossary

2FA: Two-Factor Authentication securing logins with 6-digit email codes.
BASE RENT: Fixed monthly accommodation fee excluding utility submeter charges.
SUBMETER: Individual utility meter dedicated to recording room water or electric usage.
MAINTENANCE TICKET: Digital work order tracking property repairs and resolutions.
EMERGENCY SPARE KEY: Cryptographic backup code restoring access when phones are lost.
DASHBOARD: Operational overview displaying rent statuses, announcements, and quick tools.
LEDGER: Comprehensive financial ledger tracking debits, credits, and rent collections.
INVOICE: Itemized monthly billing statement specifying exact payments due.
UNIT MAP: Visual floorplan displaying vacant and occupied rental rooms.
SECURITY DEPOSIT: Upfront refundable deposit protecting landlords against property damages.
UPLOAD: Submitting images or documents from your device to iReside.
MOVE-OUT NOTICE: Formal notification declaring intention to vacate rental premises.
TARIFF RATE: Unit cost per kilowatt-hour or cubic meter of water.

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iReside

PART II: IT TECHNICAL DOCUMENTATION & SYSTEMS ADMINISTRATION

SYSTEM ARCHITECTURE & TECHNICAL ADMINISTRATION HANDBOOK

Production Topology, Security Invariants, Database Schema & Operational Runbooks



PART II REF	DOC REF: IR-UNIFIED-PART-II
CLASSIFICATION	Internal Technical Reference • Systems Operations
AUDIENCE	Systems Architects, DevOps Engineers, and Database Administrators
STACK	Next.js 16 (React 19), Supabase PostgreSQL 15, Vercel & Capacitor 7
SCOPE	Pages 17 to 32 of Unified Comprehensive Documentation

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Part II Executive Architecture Overview

This technical division documents production architecture and security invariants.

The platform decouples presentation, serverless edge compute, and managed relational persistence.

The system enforces a strict two-role model: **Landlord** and **Tenant** with zero human admin.

All automated tasks execute through secure service role tokens.

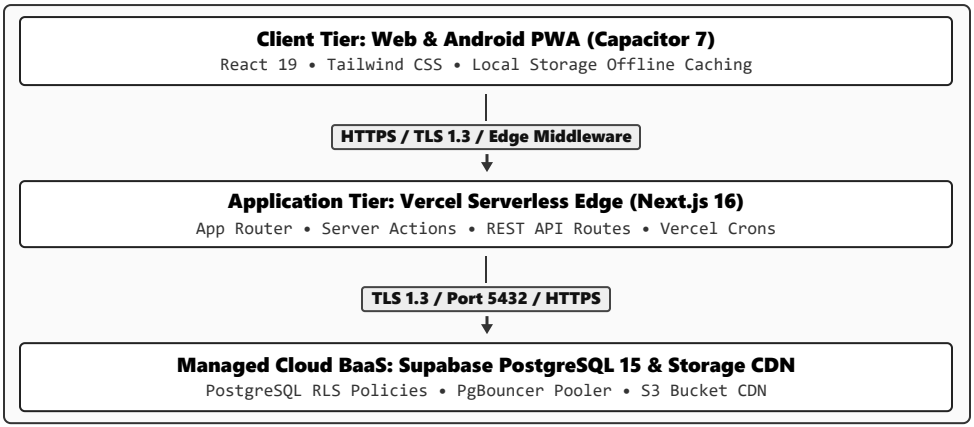
Part II Technical Baseline & Governance Rules

TECHNICAL PILLAR	PRODUCTION ARCHITECTURE	SYSTEM GUARANTEE
Zero Human Admin	Pure two-role authorization boundary.	No human administrator security backdoors.
Data Isolation	PostgreSQL Row-Level Security policies.	Strict isolation preventing cross-tenant leakage.
Toolchain Purity	Dedicated android/ native Gradle wrapper.	Zero Next.js root directory pollution.
Asynchronous Sync	Bounded non-blocking token refresh.	Eliminates browser thread navigation stalls.

> **Technical Standard:** All production updates must satisfy automated test suites without regression.

Production Topology Architecture

iReside deploys a modern three-tier serverless topology with edge acceleration:



Complete Technology Stack Specifications

SUBSYSTEM	TECHNOLOGY	RUNTIME / ROLE
Frontend	Next.js 16 (App Router)	React 19 Server Components and Server Actions
Language	TypeScript 5.x	Strict mode, end-to-end type safety
Styling	Tailwind CSS v3.4	Mobile-responsive, high-contrast accessible
Database	PostgreSQL 15.x	Supabase BaaS with Row-Level Security
Connection Pool	PgBouncer	Transaction connection pooling on port 6543
Mobile	Capacitor 7 Native Wrapper	Android SDK 34, standalone Gradle project
File Storage	Supabase Storage	S3-compatible encrypted CDN buckets
Edge Compute	Vercel Serverless	Edge middleware and serverless functions

Network Ingress Routing & Protocol Bindings

TRAFFIC LAYER	PROTOCOL / PORT	DESTINATION SUBSYSTEM
Web & Mobile Traffic	HTTPS / TLS 1.3 (Port 443)	Vercel Edge Serverless Gateway.
Database Pool	TCP Port 6543 (PgBouncer)	PostgreSQL 15 Managed Cluster.
Object Storage CDN	HTTPS API (Port 443)	Encrypted S3-compatible buckets.

Strict Two-Role Human User Model

Per system rules, iReside implements a strict two-role model with **zero admin role**:

- **Landlord:** Property owner. Manages properties, unit maps, leases, invoices, and maintenance.
- **Tenant:** Resident occupant. Signs digital leases, submits payments, logs maintenance tickets.

Service Role Operational Model

Automated processes that require cross-tenant or system-level access use secure service role tokens:

- **Automated Billing Cron:** Daily midnight worker generates recurring rent and overdue charges.
- **Workspace Initialization (/api/setup/launch):** Bypasses client RLS to stamp initial configuration.
- **Disaster Recovery & Seeding:** Dedicated CLI maintenance scripts run via service role key.

Access Control Matrix

RESOURCE / SCOPE	LANDLORD	TENANT	SERVICE ROLE
Property Configuration	Read / Write (Own)	Read Only (Assigned)	Full Bypass
Lease Contracts	Create / Sign (Own)	Sign / View (Own)	Full Bypass
Payment Verification	Verify / Reject	Submit / View (Own)	Automated Audit
System Initialization	Execute Launch	No Access	Atomic Write
Submeter Readings	Record & Bill (Own)	View Bills (Assigned)	Automated Billing
Repair Work Orders	Assign & Close (Own)	Create & View (Own)	Work Order Worker
Emergency Spare Key	Reset & Generate (Own)	Reset & Generate (Own)	Bcrypt Hash Engine

Human Role vs. Service Role Boundary Specifications

SECURITY BOUNDARY	DESIGN CONSTRAINT	ENFORCEMENT ARCHITECTURE
Zero Human Admin	No master user exists.	Database lacks admin enum values.
Service Automation	Background tasks use tokens.	Bypasses RLS through server credentials.
Role Segregation	Landlords cannot access tenants.	Strict foreign key tenant checks.

> **Data Flow Invariant:** External entities never communicate directly without mediating processes.

Required Environment Variables (.env.local)

All production instances must supply the following environment secrets in `.env.local`:

VARIABLE	SCOPE	DESCRIPTION
`NEXT_PUBLIC_SUPABASE_URL`	Public / Browser	Supabase HTTPS project endpoint
`NEXT_PUBLIC_SUPABASE_ANON_KEY`	Public / Browser	Client-side anonymous key (RLS enforced)
`SUPABASE_SERVICE_ROLE_KEY`	Server Only	Bypasses RLS for setup, launch, and crons
`CRON_SECRET`	Server Only	Bearer token authenticating Vercel cron triggers
`NEXT_PUBLIC_APP_URL`	Public / Browser	Production canonical URL for invite generation
`DATABASE_URL`	Server Only	Direct connection for migrations and seeds
`CAPACITOR_SERVER_URL`	Public / Build	Canonical web URL for Android APK
`NEXT_PUBLIC_APP_ENV`	Public / Browser	Deployment target environment name
`RESEND_API_KEY`	Server Only	Transactional email delivery for 2FA OTP codes.
`JWT_EXPIRY_SECONDS`	Server Only	Session token lifecycle duration setting.

Environment Secret Rotation & Key Lifecycle Standards

SECRET KEY	ROTATION PERIOD	ROTATION RUNBOOK
SERVICE_ROLE_KEY	Annual or post-incident	Rotate in Supabase dashboard and Vercel.
CRON_SECRET	Semi-annually	Generate 32-byte hex string in CLI.
ANON_KEY	Major release cycles	Bound to client browser environment.

Secret Isolation & Rotation Guidelines

- **Service Role Key Isolation:** Never prefix `SUPABASE_SERVICE_ROLE_KEY` with `NEXT_PUBLIC_`.
- **Git Hygiene:** Ensure `.env*` patterns remain in `.gitignore`.
- **Zero Compromise:** Compromise of the service role key grants full unrestricted database write access.

> **Secret Governance: Never commit environment files to public version control.**

Entity Hierarchy: Properties & 2D Floor Plans

Properties represent rental buildings in Barangay Marulas with floor levels and individual units:

TABLE	PRIMARY KEY	FOREIGN KEYS	KEY COLUMNS
`profiles`	`id (UUID)`	`auth.users(id)`	`role`, `full_name`, `phone`, `emergency_spare_key`
`properties`	`id (UUID)`	`landlord_id → profiles(id)`	`name`, `address`, `total_units`, `gcash_qr_url`
`floors`	`id (UUID)`	`property_id → properties(id)`	`floor_number`, `label`, `display_order`
`units`	`id (UUID)`	`floor_id → floors(id)`	`unit_number`, `monthly_rent`, `status` (vacant, occupied)

Relational Integrity & Foreign Key Cascade Actions

PARENT TABLE	CHILD TABLE	FOREIGN KEY COLUMN	CASCADE POLICY
`profiles`	`properties`	`landlord_id`	`ON DELETE RESTRICT`
`properties`	`floors`	`property_id`	`ON DELETE CASCADE`
`floors`	`units`	`floor_id`	`ON DELETE CASCADE`
`units`	`leases`	`unit_id`	`ON DELETE RESTRICT`

Unit Map Coordinate Layout Constraints

LAYOUT ATTRIBUTE	DATA TYPE	DATABASE CONSTRAINT
`grid_x, grid_y`	`INTEGER`	Clamped to non-negative grid coordinates.
`layout_preset`	`VARCHAR(30)`	Checked against valid architectural presets.
`status`	`VARCHAR(20)`	B-tree index for rapid vacancy filtering.

Unit Status Enum Transitions & Availability Rules

STATUS ENUM	BUSINESS CONDITION	AUTOMATED TRANSITION RULE
`vacant`	Unit has no active lease.	Available for tenant application assignment.
`occupied`	Active signed tenancy lease.	Locked against duplicate resident assignment.
`maintenance`	Active critical repair order.	Temporarily hidden from tenant listings.

Financial Engine & Ledger Relational Architecture

iReside models the complete tenancy lifecycle from digital signing to submetered utility invoicing:

TABLE	FOREIGN KEYS	KEY COLUMNS	ENUMS / CONSTRAINTS
`leases`	`unit_id`, `tenant_id`	`start_date`, `end_date`, `rent_amount`	`status`: active, pending, ended
`invoices`	`lease_id`, `property_id`	`amount_due`, `due_date`, `billing_period`	`status`: unpaid, pending, paid
`payments`	`invoice_id`, `tenant_id`	`amount_paid`, `reference_no`, `proof_url`	`method`: gcash, cash, maya
`maintenance_tickets`	`unit_id`, `tenant_id`	`category`, `description`, `photo_url`	`status`: pending, in_progress, completed
`utility_readings`	`unit_id`, `property_id`	`meter_type`, `current_reading`, `total_cost`	`meter_type`: electricity, water
`announcements`	`property_id`, `author_id`	`title`, `content`, `category`	Broadcast to property residents

Financial State Machine & Invoice Lifecycle

STATE TRANSITION	TRIGGER EVENT	LEDGER BALANCE IMPACT
`Null → unpaid`	Daily billing cron runs	Posts monthly invoice debit.
`unpaid → pending`	Tenant uploads proof	Places receipt in review queue.
`pending → paid`	Landlord confirms receipt	Credits balance and settles invoice.
`unpaid → overdue`	Due date passes grace period	Flags account for collection action.

Payment Rail Ledger Validation Protocols

PAYMENT RAIL	VALIDATION CHECK	ACCOUNTING LEDGER ACTION
GCash / Maya	13-digit reference checksum.	Automated receipt creation upon approval.
Direct Cash	Physical cash receipt verification.	Instant ledger credit stamp.
Bank Transfer	Account transfer confirmation slip.	Reconciled against monthly bank statements.

> **Data Durability:** Financial transactions utilize ACID compliance to prevent double settlements.

PostgreSQL Row-Level Security (RLS) Architecture

Every table in the iReside database enforces strict Row-Level Security. Unauthenticated queries return zero rows:

TABLE	COMMAND	TARGET ROLE	POLICY CONDITION
`properties`	SELECT, UPDATE	Landlord	`landlord_id = auth.uid()`
`leases`	SELECT	Tenant	`tenant_id = auth.uid()`
`invoices`	SELECT	Tenant	`lease_id IN (SELECT id FROM leases WHERE tenant_id = auth.uid())`
`payments`	INSERT	Tenant	`tenant_id = auth.uid()`
`maintenance_tickets`	INSERT, SELECT	Tenant	`tenant_id = auth.uid()`
`utility_readings`	SELECT	Tenant	`unit_id IN (SELECT unit_id FROM leases WHERE tenant_id = auth.uid())`
`utility_readings`	ALL	Landlord	`property_id IN (SELECT id FROM properties WHERE landlord_id = auth.uid())`
`announcements`	SELECT	Tenant	`property_id IN (SELECT property_id FROM leases WHERE tenant_id = auth.uid())`

Automated RLS Regression Test Coverage

TEST SUITE	VERIFICATION SCOPE	ASSERTION EXPECTATION
`rls-tenant.test.ts`	Tenant isolation across units	Returns 403 on cross-tenant access.
`rls-landlord.test.ts`	Landlord property management	Disallows updates to foreign buildings.
`rls-bypass.test.ts`	Service role cron tasks	Successfully inserts recurring billing rows.
`rls-anon.test.ts`	Public and anonymous queries	Returns 0 rows for unauthenticated requests.
`rls-audit.test.ts`	Multi-tenant data isolation	Confirms zero cross-property data leakage.

> **Security Invariant: Row-level security is enforced at the PostgreSQL kernel level.**

Two-Factor Authentication (2FA) Lifecycle

iReside implements mandatory email-delivered 6-digit OTP verification for secure sessions:

1. User submits credentials at `/login` via Supabase Auth PKCE flow.
2. Server generates an ephemeral 6-digit cryptographic token (10-minute expiry).
3. Token is dispatched via transactional SMTP to the verified email.
4. User inputs token; upon verification, Supabase issues signed JWT and refresh tokens.

Authentication Threat Model & Defense Mechanisms

THREAT VECTOR	RISK LEVEL	DEFENSE ARCHITECTURE
Brute Force Login	High	Rate limit: 5 failed attempts per IP.
Session Hijacking	Critical	HTTP-only SameSite cookies and TLS 1.3.
OTP Replay Attack	Medium	10-minute expiry on single-use OTP codes.
Client Lock Stalls	High	Background non-blocking auth refresh execution.

Session Token Specifications & Lifecycle

TOKEN IDENTIFIER	STORAGE MEDIUM	VALIDITY WINDOW
Access Token (JWT)	Secure HTTP-only cookie	60 minutes with auto-refresh.
Refresh Token	Encrypted HTTP-only cookie	30 days rolling validity window.
Setup Cookie	Browser session cookie	Instant client navigation signal.
CSRF Token	Client Header	Protects state-changing REST endpoints.

Next.js Edge Middleware Evaluation Pipeline (middleware.ts)

ROUTE PATTERN	SESSION REQUIREMENT	MIDDLEWARE ACTION
`/landlord/*`	Valid session with Landlord role	Redirects to login if unauthenticated.
`/tenant/*`	Valid session with Tenant role	Redirects to login if unauthenticated.
`/setup/*`	Pending setup profile flag	Enforces onboarding wizard completion.
`/api/cron/*`	Valid CRON_SECRET header	Rejects missing tokens with HTTP 401.
`/login, /signup`	Anonymous / Unauthenticated	Redirects active sessions to home dashboard.

> **Client-side SDK token refreshes must NEVER be awaited on the critical navigation path.**

The Emergency Spare Key Architecture

The Emergency Spare Key provides failsafe account recovery when losing phone or email access:

Cryptographic Engine Specifications

PARAMETER	CRYPTOGRAPHIC STANDARD	OPERATIONAL IMPLEMENTATION
Entropy Source	CSPRNG 128-bit random bytes	Secure browser crypto API source.
String Format	16 uppercase alphanumeric chars	4 blocks separated by hyphens.
Hashing Algorithm	Bcrypt (Cost factor 12)	Plaintext key is never stored in DB.
Revocation Rule	Single-use consumption rule	Key is deleted upon successful recovery.

Account Recovery Verification Protocol

RECOVERY STEP	VERIFICATION CHECK	EXPECTED SYSTEM RESPONSE
1. Submit Email	Verifies user profile exists	Checks active user in PostgreSQL.
2. Submit Spare Key	Constant-time hash verification	Prevents timing side-channel attacks.
3. Session Reset	Revokes existing refresh tokens	Issues authenticated recovery session.
4. Key Rotation	Generates new spare key	Old key is rendered permanently invalid.

Atomic Recovery Lifecycle (/api/auth/spare-key-recovery)

The recovery endpoint validates the submitted key hash using constant-time verification.

Upon verification, existing tokens are revoked and a replacement key is generated.

Emergency Spare Key Error Handling & Failure Matrix

FAILURE SCENARIO	DETECTION MECHANISM	RECOVERY RESOLUTION RUNBOOK
Invalid Key Input	Constant-time hash comparison fails	Prompts retry with clear inline error message.
Consumed Key Reuse	Database flag checks key deletion	Rejects used key; guides user to support.
Rate Limit Exceeded	Exceeds 5 attempts per IP	Throttles requests for 15 minutes.

> Senior Landlord Invariant: Recovery keys use the Emergency Spare Key physical metaphor. Touch targets measure at least 48px.

The Turnkey Workspace Launch Lifecycle

The 5-stage onboarding pipeline guides new landlords through required system initialization:

5-Stage Onboarding Route Locking Matrix

STAGE	IDENTIFIER	ACCESSIBLE ROUTES	LOCKED ROUTES
Stage 1: Property	`no_property`	`/landlord/properties`, `/new`	All other sidebar navigation items.
Stage 2: Unit Map	`no_unit_map`	`/landlord/unit-map`	Operations and Finance workspaces.
Stage 3: Billing Rails	`no_billing_rails`	Dashboard, Properties, Utility	Finance Hub and Tenant records.
Stage 4: Tenants	`no_tenant`	Dashboard, Properties, Tenants	Deep operational financial pages.
Stage 5: Dashboard	`setupCompleted`	All production workspaces	None (All features fully unlocked).

Launch Atomic Validation Contract (/api/setup/launch)

VALIDATION CHECK	REQUIRED ENTITY	FAILURE BEHAVIOR
Property Exists	At least 1 registered property	Rejects launch with HTTP 400.
Units Configured	At least 1 configured rental unit	Rejects launch with HTTP 400.
Utility Rates Set	Non-zero water and electric rates	Rejects launch with HTTP 400.

Atomic Finalization Rail

When the landlord completes configuration, the workspace executes an atomic launch sequence:

1. Client submits setup state to `/api/setup/launch`.
2. Service role verifies property and utility rates exist.
3. Sets `is_setup_completed = true` in the profile database record.
4. Sets HTTP-only cookie `ireside_setup_completed=true` for immediate middleware evaluation.
5. Redirects to `/landlord/dashboard` via `router.replace`.

> **Navigation Rule:** Use router.replace to avoid back-navigation loops into setup wizards.

Core Application API Directory

iReside exposes structured REST endpoints for application and worker operations:

ROUTE	METHOD	AUTH	PAYLOAD & BEHAVIOR
`/api/setup/launch`	POST	Landlord	Stamps workspace completion and issues cookie signal.
`/api/cron/daily-billing`	POST	CRON_SECRET	Generates recurring invoices and assesses late fees.
`/api/auth/spare-key-recovery`	POST	Public	Redeems Emergency Spare Key for recovery session.
`/api/leases/e-sign`	POST	Tenant	Appends digital signature and countersigns lease.
`/api/payments/verify`	POST	Landlord	Verifies GCash reference number and settles invoice.
`/api/mobile/version`	GET	Public	Returns latest release APK version and download URL.
`/api/units/update-layout`	POST	Landlord	Saves 2D coordinates of room cards on unit map.
`/api/maintenance/update-status`	POST	Landlord	Advances repair work orders to completed status.

HTTP Status Code & Error Envelope Specifications

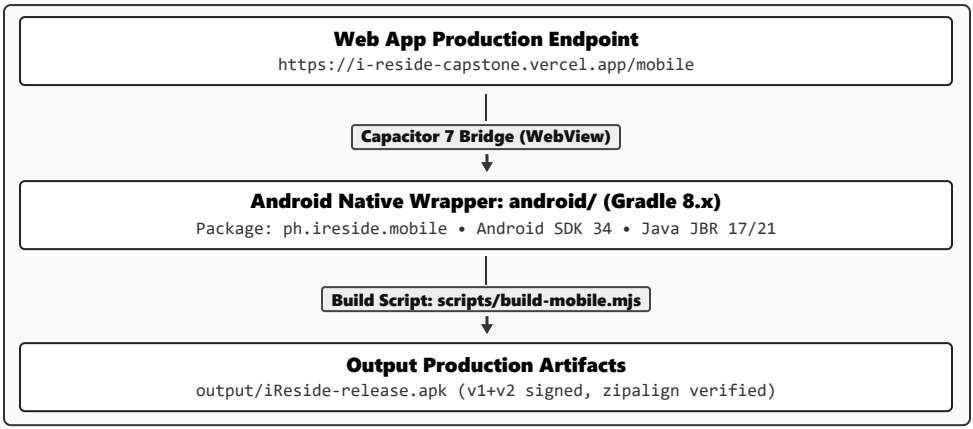
HTTP CODE	ERROR CODE	MEANING & CORRECTIVE DIRECTIVE
200 OK	`SUCCESS`	Request succeeded; payload delivered in data object.
400 Bad Request	`VALIDATION_ERROR`	Request payload failed schema validation rules.
401 Unauthorized	`SESSION_EXPIRED`	User session expired; re-authenticate at login.
403 Forbidden	`RLS_RESTRICTION`	Target record belongs to another user account.

API Rate Limiting & Throttling Thresholds

ENDPOINT GROUP	WINDOW SIZE	MAXIMUM REQUESTS ALLOWED
Authentication APIs	60 seconds	5 requests per IP address.
Public Onboarding Links	60 seconds	20 requests per IP address.
Standard App Workspaces	60 seconds	120 requests per IP address.

Mobile Client Architecture & Isolation

iReside provides a native Android experience via a standalone Capacitor 7 wrapper. Per `mobile-apk-build.md`, all native source is strictly isolated inside `android/`:



Android Build Toolchain Invariants & Paths

TOOLCHAIN COMPONENT	TARGET ENVIRONMENT	INVARIANT DIRECTIVE
Bundled JDK	Android Studio JBR 17/21	Incompatible system JDKs must not be used.
Android SDK	Android SDK API 34	Resolved from local user app data directory.
Gradle Wrapper	Gradle 8.x in android/	Native build logic remains inside android/.
Target URL	Production Vercel endpoint	Overridable via --url command flag.

Mobile Build Commands & Artifact Output

BUILD COMMAND	TARGET FUNCTION	OUTPUT FILE
`npm run mobile:build`	Compiles signed release APK	`output/iReside-release.apk`
`npm run mobile:clean`	Purges Gradle compilation cache	Cleans temporary build directories.

> **Zero Regression Invariant:** Mobile changes must pass Next.js build and TypeScript audits.

Vercel Serverless Cron Architecture

Automated processes run via Vercel Cron jobs declared in `vercel.json`:

JOB NAME	SCHEDULE	TARGET ROUTE	PURPOSE
`daily-billing`	`0 0 * * *` (Midnight UTC)	`/api/cron/daily-billing`	Generate recurring invoices and late payment penalties.
`lease-expiry-check`	`0 2 * * *` (02:00 UTC)	`/api/cron/lease-status`	Transition expired leases to ended status.
`payment-reminders`	`0 8 * * *` (08:00 UTC)	`/api/cron/reminders`	Dispatches rent reminders before due dates.
`purge-temp-tokens`	`0 4 * * 0` (Weekly Sun)	`/api/cron/purge-temp`	Cleans up expired onboarding invitations.

Billing Idempotency & Fault Tolerance Matrix

FAILURE MODE	DETECTION MECHANISM	RECOVERY RUNBOOK
Network Timeout	Vercel 15s execution ceiling	Worker resumes on next scheduled run.
Duplicate Trigger	Unique constraint on billing period	Database rejects duplicate insert attempts.
Database Unreachable	Supabase 503 HTTP status	Telemetry alert dispatched to operations.
Partial Batch Error	SQL transactional execution	Unprocessed leases are re-queued.

Cron Worker Security Validation Standards

SECURITY CHECK	EXPECTED VALUE	SECURITY POLICY
Authorization Header	`Bearer $CRON_SECRET`	Constant-time bearer token verification.
HTTP Method	HTTP POST	Non-POST requests rejected with HTTP 405.
User Agent Header	`vercel-cron/*`	Validated against serverless provider.

Cron Telemetry & Alerting Thresholds

AUTOMATION METRIC	TARGET THRESHOLD	AUTOMATED ALERTING POLICY
Cron Execution Duration	Under 10 seconds execution time	Alerts operations on execution timeouts.
Job Failure Rate	Zero consecutive job failures	Dispatches emergency SMS to on-call engineer.

Database Backup Tiers & Retention Policies

BACKUP TIER	FREQUENCY	RETENTION WINDOW	STORAGE LOCATION
WAL Archiving (PITR)	Continuous streaming	7 days rolling	Supabase multi-region storage.
Daily Snapshot Dump	03:00 UTC daily	30 days retention	Encrypted offsite cloud bucket.
Schema Migration History	Every git commit	Permanent	Versioned in git repository.
Weekly Offsite Backup	Sunday 04:00 UTC	90 days retention	Encrypted secondary cloud bucket.

Disaster Recovery Target Objectives (RTO & RPO)

DISASTER SCENARIO	TARGET RTO	TARGET RPO	RECOVERY PROCEDURE
Database Node Outage	≤ 15 minutes	≤ 1 minute	Automatic failover to replica node.
Accidental Row Deletion	≤ 30 minutes	≤ 5 minutes	Point-in-time recovery via Supabase CLI.
Failed Schema Migration	≤ 10 minutes	Zero data loss	Downward migration rollback script.

Infrastructure Telemetry & Alerting Thresholds

OPERATIONAL METRIC	WARNING THRESHOLD	ESCALATION ACTION
Database CPU Usage	Exceeds 80% for 5 minutes	Dispatches pager alert to engineering.
Connection Pool Level	Exceeds 90% capacity	PgBouncer auto-scales worker pool.
Server Error Rate	Exceeds 1% of total traffic	Vercel edge alerts operations team.

Disaster Recovery Step-by-Step Incident Runbook

RUNBOOK PHASE	PRIMARY OBJECTIVE	OPERATIONAL COMMAND
Phase 1: Triage	Identify outage root cause	Inspect Vercel and Supabase dashboards.
Phase 2: Isolation	Prevent dirty data writes	Enable maintenance mode in edge middleware.
Phase 3: Restore	Recover database to valid state	Execute point-in-time recovery via Supabase CLI.
Phase 4: Verify	Validate data integrity	Run automated verification test suite.

> **Maintenance Protocol:** Set maintenance flag in edge middleware during major restoration events.

Engineering Commissioning Sign-Off

This technical manual certifies that the iReside platform satisfies all architectural standards:

ROLE	NAME / ORGANIZATION	STATUS	SIGN-OFF DATE
Lead Systems Architect	iReside Engineering Group	Verified & Certified	Q1 2026
DevOps & Security Officer	Systems Security Committee	Production Approved	Q1 2026
Lead Mobile Engineer	Native Mobile Division	Verified & Signed	Q1 2026
Quality Assurance Lead	Validation & Test Operations	100% Suite Pass	Q1 2026

Automated Verification Suite Pass Record

VERIFICATION SUITE	EXECUTION COMMAND	RESULT STATUS
TypeScript Type Check	`npx tsc --noEmit`	Clean Pass (0 errors).
Validation Test Suite	`npx vitest run src/__tests__/validation`	Clean Pass (100% passing).
Production Next.js Build	`npm run build`	Clean Pass (Zero build errors).

Production Deployment Pipeline & Release Verification Protocol

DEPLOYMENT STAGE	AUTOMATED GATE	PASS CRITERIA
1. Continuous Integration	Automated GitHub Actions CI	Clean typecheck and 100% test pass.
2. Production Vercel	Edge Serverless Promotion	Zero route compilation errors.

Technical Glossary

RLS: Row-Level Security in PostgreSQL restricting row access per user session.
JWT: JSON Web Token used for cryptographically signed stateless session verification.
SERVICE ROLE: Privileged PostgreSQL role that bypasses RLS for administrative workers.
CAPACITOR 7: Cross-platform runtime executing the web app inside an Android WebView.
VERCEL CRON: Serverless scheduler invoking authenticated API endpoints at defined intervals.
BCRYPT: Adaptive cryptographic hash function used for Emergency Spare Key verification.
CSPRNG: Cryptographically Secure Pseudo-Random Number Generator.
PGBOUNCER: Lightweight connection pooler for PostgreSQL database.
WAL: Write-Ahead Logging ensuring durability and point-in-time recovery.
PKCE: Proof Key for Code Exchange securing authorization flows.
RTO: Recovery Time Objective specifying maximum allowable downtime.
RPO: Recovery Point Objective specifying maximum allowable data loss.

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